

Phishing Email Detection & Awareness System

CYBERSECURITY ASSESSMENT & TECHNICAL MATRIX REPORT

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Task Objective:	Analyze email headers, evaluate message authentication parameters (SPF/DKIM), classify risk properties, and deliver non-technical user awareness recommendations.
Document Type:	Technical Evaluation Deliverable (Task 2)
Platform Architecture:	Canva Visual Layout Master Layout & Public GitHub Repository Documentation
Environment Scope:	Google Admin Toolbox Messageheader Analyzer Configuration

Technical Analysis & Core Findings

1. Executive Summary

Modern workplace communications face constant evaluation threats from advanced corporate infrastructure impersonations. This project provides a step-by-step structural investigation into standard communication vectors by analyzing text templates using cryptographic and protocol verification rules.

Using the Google Admin Toolbox Messageheader validation interface, three distinct testing profiles were examined. The outcomes compiled in this matrix showcase that baseline protocol verification (SPF and DKIM) remains the initial line of infrastructure protection, yet must be reinforced by continuous psychological awareness strategies to offset high-risk social engineering attempts.

2. System Assessment Matrix

This systematic summary cross-references the verified infrastructure profiles evaluated through our toolbox routine:

Artifact ID	Declared Identity	Technical Parameter Findings	Risk Classification
Sample 01	Microsoft IT Helpdesk	- SPF Protocol Check: FAIL - DKIM Check: FAIL - Host Domain: badactorinfrastructure.net	 PHISHING ATTACK
Sample 02	Global Rewards Club	- SPF Protocol Check: PASS - DKIM Check: PASS - Payload Element: Legacy .doc file block	 SUSPICIOUS CONTENT
Sample 03	Corporate HR Dept	- SPF Protocol Check: PASS - DKIM Check: PASS - Core Metrics: Domain and signature fully verified	 SAFE BASELINE

3. Deep-Dive Sample Investigations

Case Study Profile 01: Critical Password Expiry Deception

Vector Profile: Brand Impersonation and Credential Harvesting.

The analytical verification environment explicitly flagged structural domain authentication failures. The sending server IP failed the SPF verification protocol, indicating that it possessed no legitimate credentials to communicate as an agent for Microsoft.

Google Admin Toolbox Messageheader

Help

MessageID	EXPIRED-PASSWORD-ALERT-9921831@corporate.com
Created at:	6/3/2023, 4:24:00 PM GMT+5:30 (Delivered after 12 hours)
From:	"Microsoft IT Helpdesk" <alerts@microsoftsecurityupdate.com>
To:	employee@corporate.com
Subject:	CRITICAL: Password Expiry Notice - Immediate Action Required
SPF:	fail with IP Unknownt Learn more
DKIM:	fail with domain Unknownt Learn more

#	Delay	From *	To *	Protocol	Time received
1	12 hours	badactor@infrastructure.net	→ [Google] ms.google.com	ESMTPS	6/4/2023, 4:54:00 AM GMT+5:30

ANALYZE ANOTHER HEADER

Business-Friendly Remediation Context: This attack plays on an employee's professional anxieties by enforcing a fabricated 4-hour window. Non-technical staff must recognize that genuine IT administration teams do not distribute unencrypted HTTP baseline links for core authentication alterations.

Case Study Profile 02: High-Value Financial Incentives

Vector Profile: Malicious Payload Delivery and Financial Baiting.

A high-integrity structural challenge: the message safely passes baseline network-layer protocols (SPF/DKIM) because it uses legitimate commercial distribution environments. The primary danger vector resides entirely within the attached unstructured file layer.

Google Admin Toolbox

Messageheader

Help

MessageID	REWARDS-CLAIM-7721305@external-marketing-vendor.com				
Created at:	6/3/2023, 4:26:00 PM GMT+5:30 (Delivered after 12 hours)				
From:	"Global Rewards Club" <rewards@external-marketing-vendor.com>				
To:	employee@corporate.com				
Subject:	CONGRATULATIONS! Claim your \$500 Amazon Gift Card Now!				
SPF:	pass with IP 203.0.113.88 Learn more				
DKIM:	pass with domain Unknownt Learn more				

#	Delay	From *	To *	Protocol	Time received
1	12 hours	marketingcloud.com	→ Google! ms.google.com	ESMTPS	6/4/2023, 4:56:00 AM GMT+5:30

ANALYZE ANOTHER HEADER

Business-Friendly Remediation Context: Attackers exploit behavioral positive reinforcement matrices by deploying unexpected reward messaging. The attached document is weaponized to trigger background script execution blocks. Staff must understand that valid accounting elements never utilize legacy macro-enabled blocks to process incentives.

Case Study Profile 03: Standard Open Enrollment Verification

Vector Profile: Secure Operational Standard Reference.

The technical metadata perfectly matches the organizational source. The network path matches, alignments resolve cleanly, and the text uses clear, non-aggressive communication paths.

Google Admin Toolbox Messageheader

Help

MessageId: HR-ENROLLMENT-2026-112233@internal-hr.com

Created at: 6/3/2026, 4:27:00 PM GMT+5:30 (Delivered after 12 hours)

From: "Corporate HR Department" <notifications@internal-hr.com>

To: employee@corporate.com

Subject: Reminder: Open Enrollment Period for Health Insurance Ends Next Friday

SFV: pass with IP 192.0.2.15
[Learn more](#)

DKIM: pass with domain Unknown! [Learn more](#)

#	Delay	From *	To *	Protocol	Time received
1	12 hours	internal-hr.com	→  mx.google.com	SMTPS	6/4/2026, 4:57:00 AM GMT+5:30

ANALYZE ANOTHER HEADER

4. Corporate User Awareness Guidelines

To prevent baseline security compromises, organizations should enforce the structured **"STOP" Verification System** for processing every incoming mail asset:

- **S - Scrutinize Infrastructure Addresses:** Look directly past the friendly text title strings. Inspect the domain details hosted within the right-angle brackets <>. If character variations or typosquat structures exist, freeze interaction.
- **T - Trace Route Links Safely:** Hover the system pointer over buttons or active lines to review destination paths without executing clicks. Look out for unencrypted protocols.
- **O - Observe Context Patterns:** Treat any message with suspicion if it establishes urgent artificial expiration terms, requires unusual password confirmations, or contains unprompted files.
- **P - Protect via Operational Reporting:** Do not copy, distribute, or forward detected hazards to immediate team members. Route the raw document blocks directly to the designated Threat Incident Management mailbox.

Operational Submission Note

This technical verification project, containing all raw code components and live analytical documentation matrices, has been successfully compiled and distributed via our public GitHub code management portal for immediate audit.